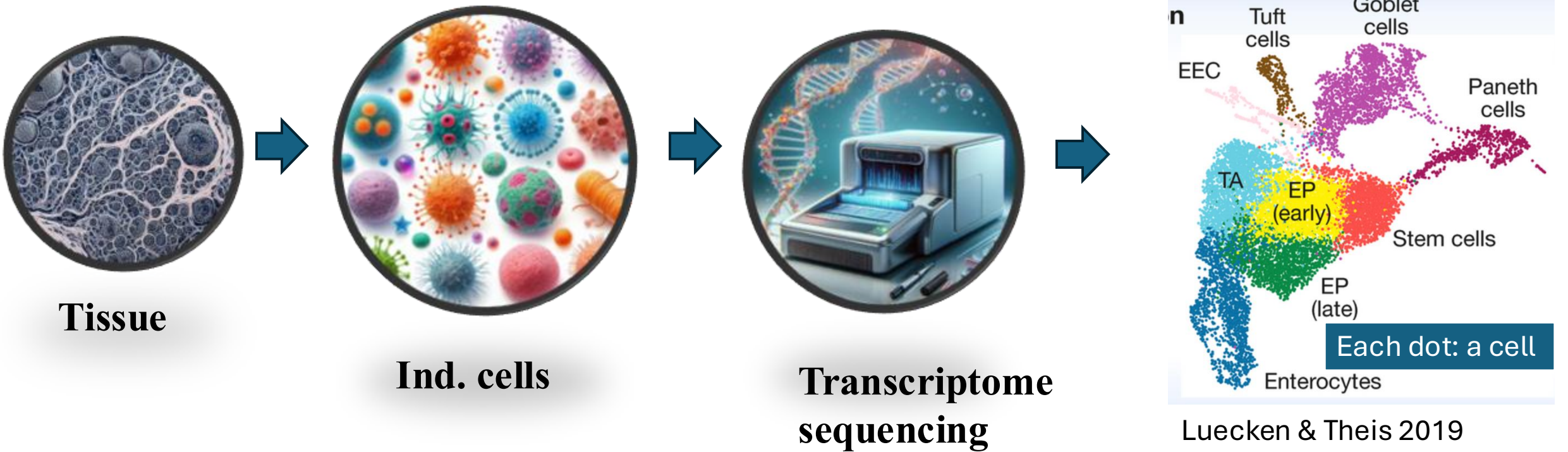
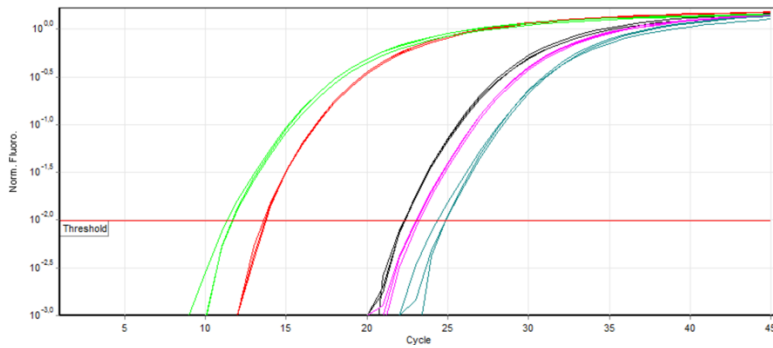


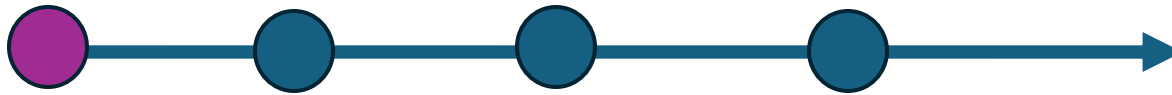
Single cell transcriptomics/ single cell RNA sequencing/ scRNAseq



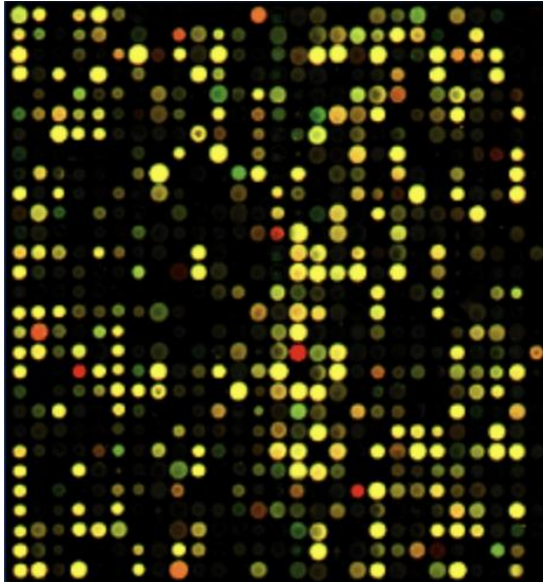
qRT-PCR



- PCR
- Sensitive
- A small number of genes/transcripts

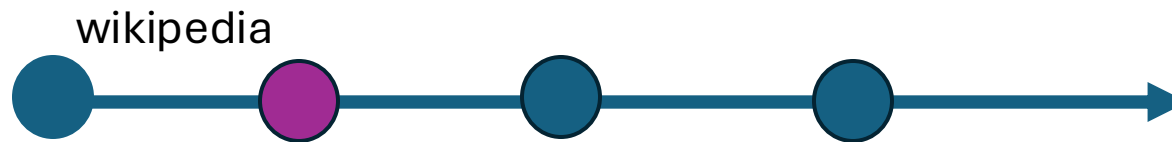


microarray

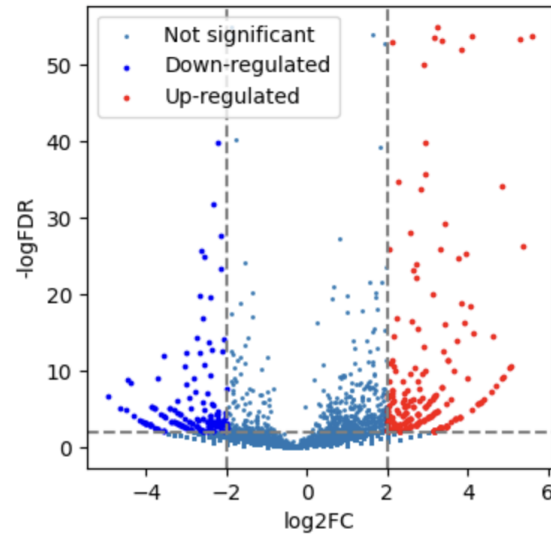


- Hybridization based
- Genome wide survey of the **transcriptome**
- Only profile predefined transcripts/genes on the chip

A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

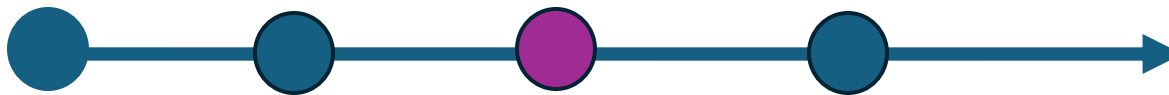


Bulk RNAseq

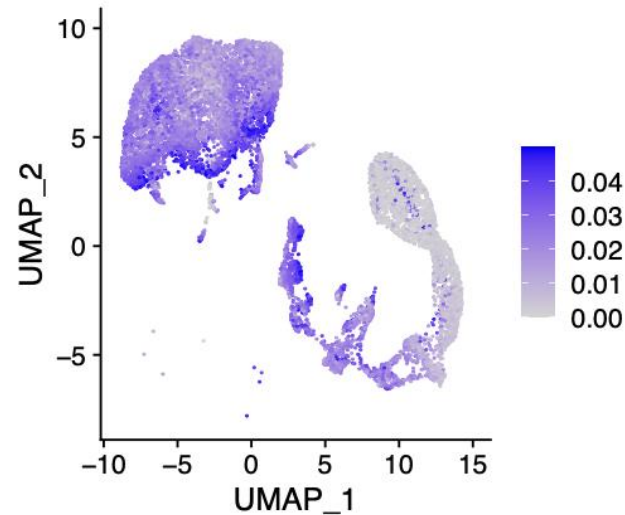


A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

- Full sequencing of the whole **transcriptome**



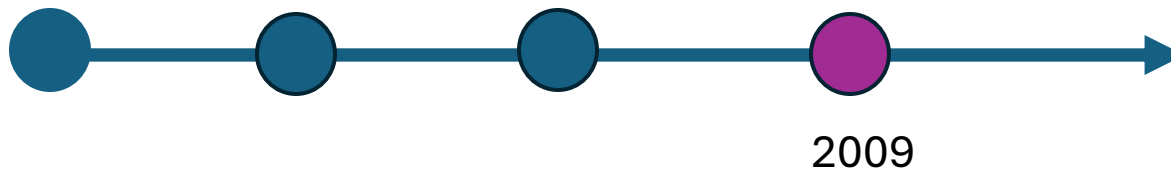
scRNAseq

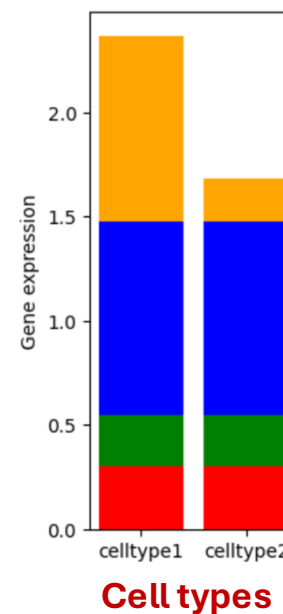
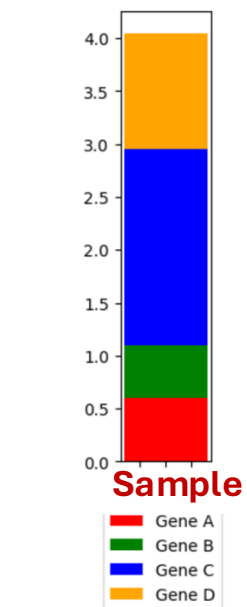
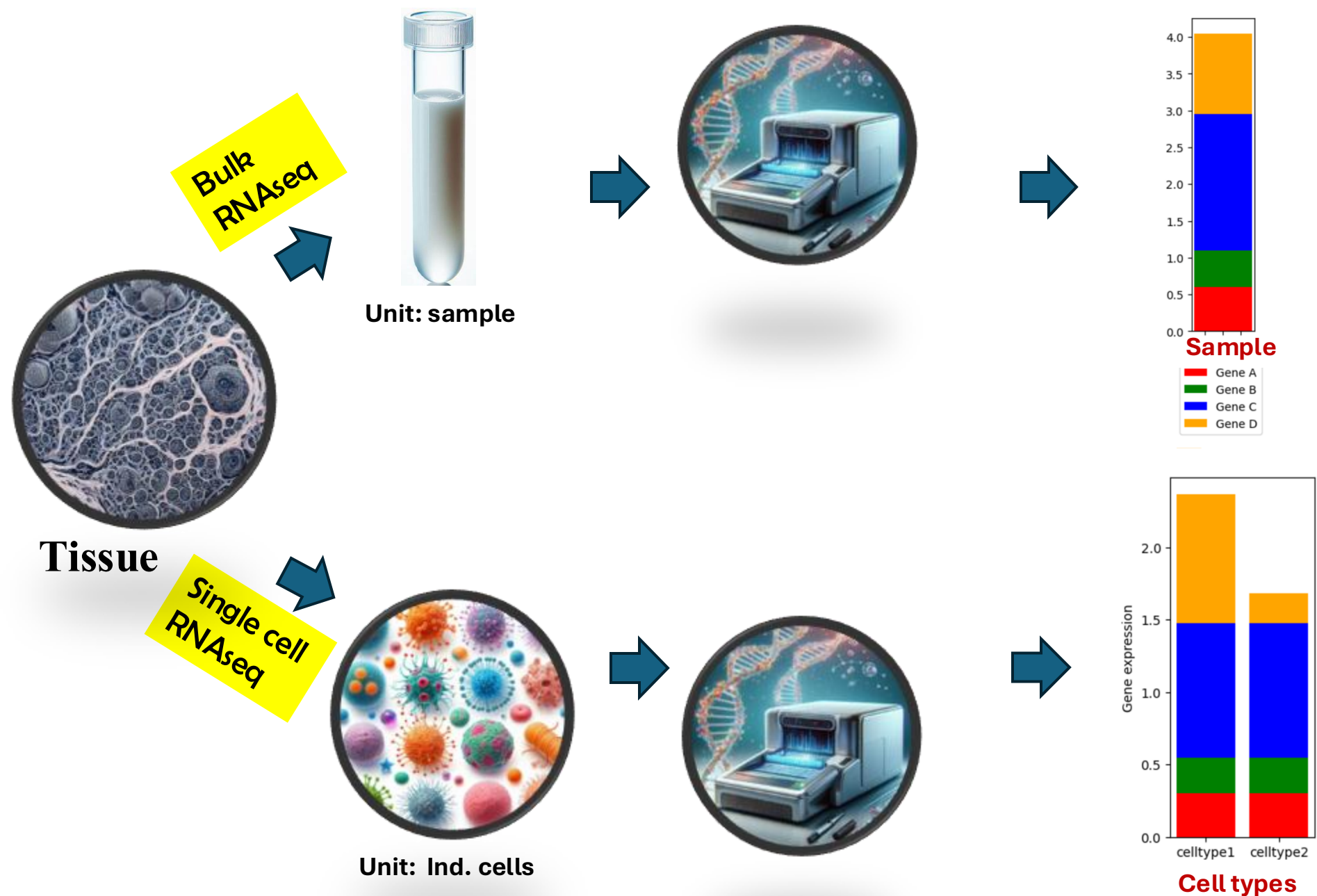


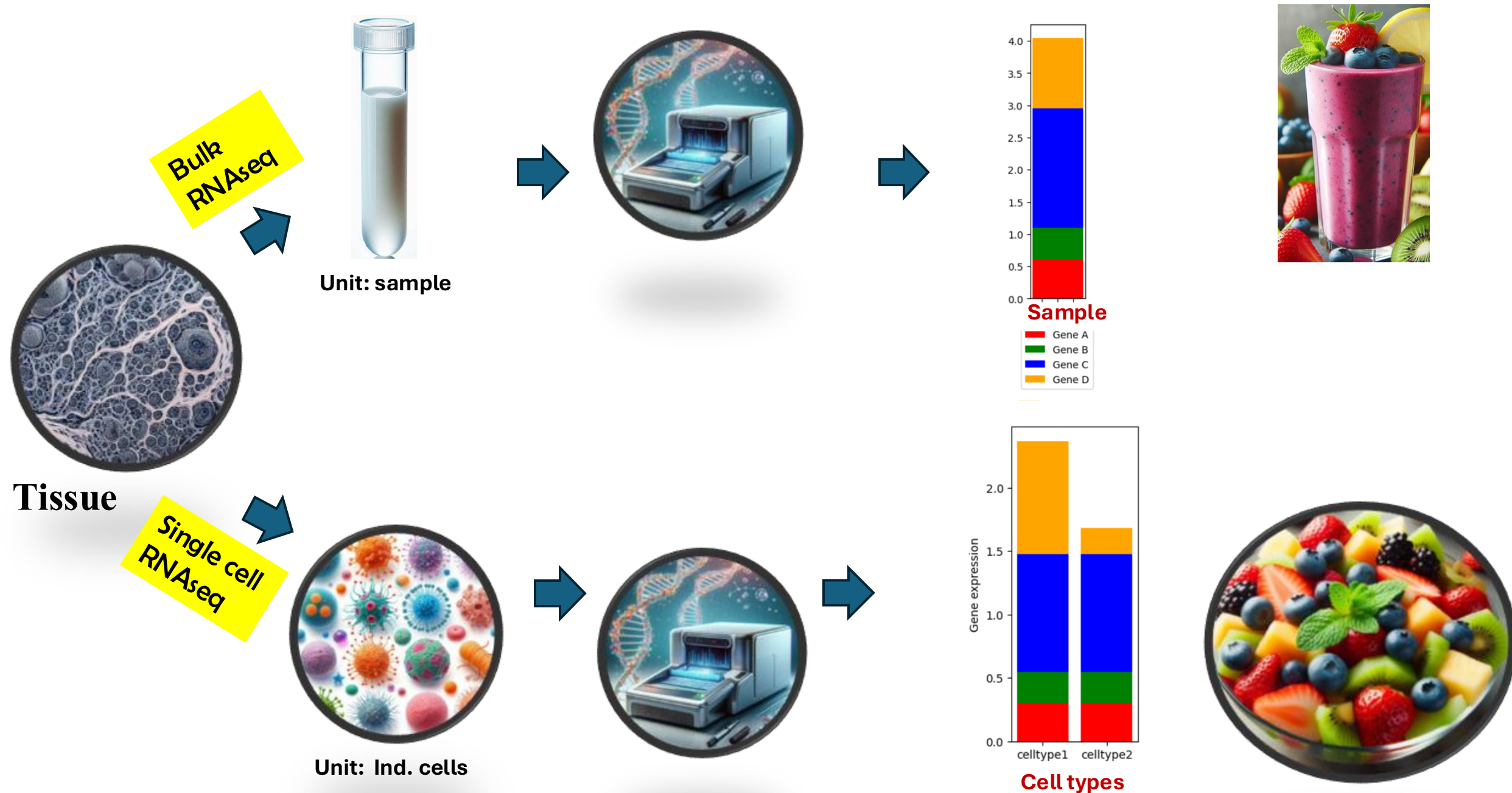
- Sequencing
- Genome wide survey of the **transcriptome**
- Individual cells

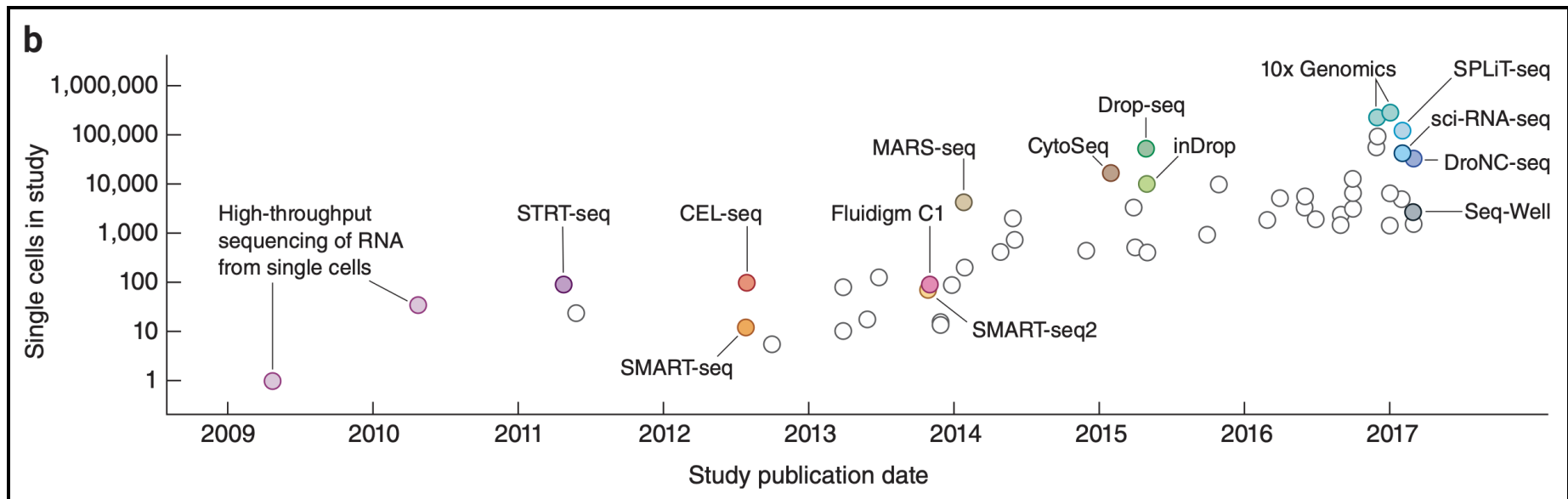


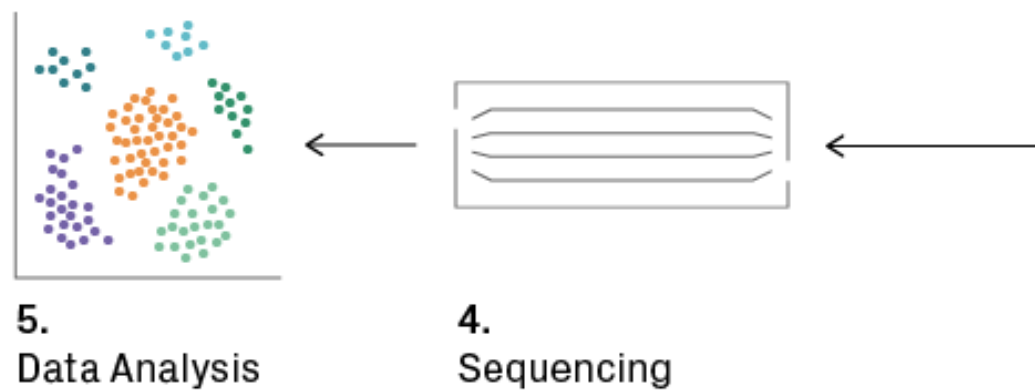
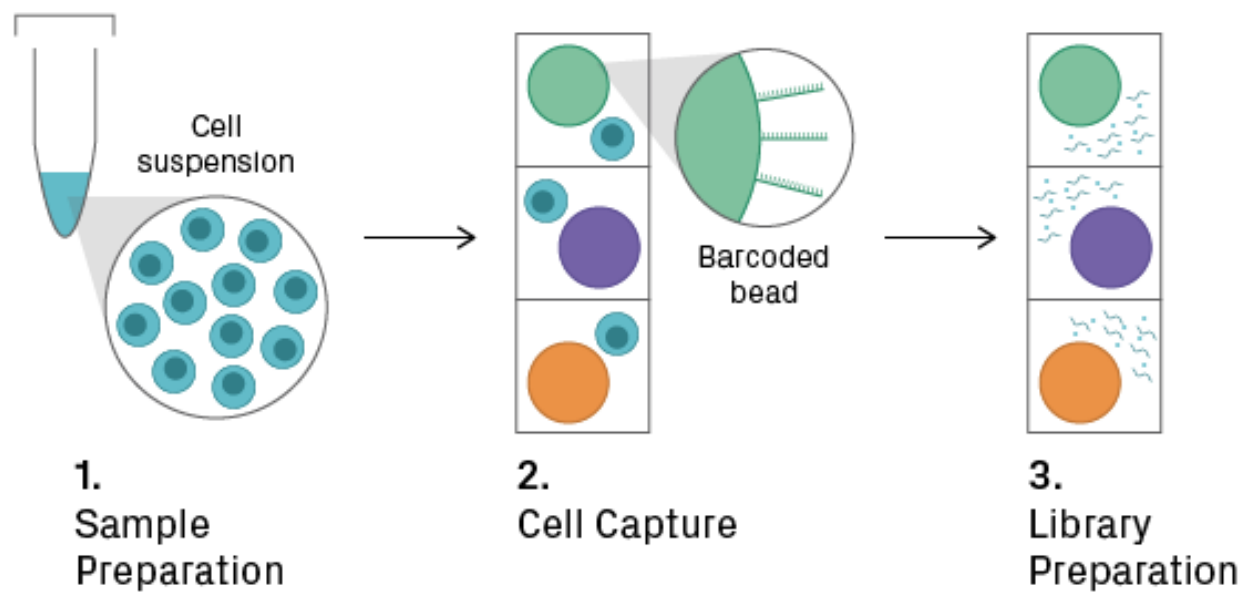
A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

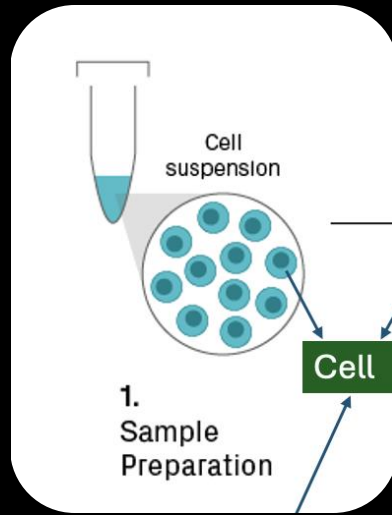


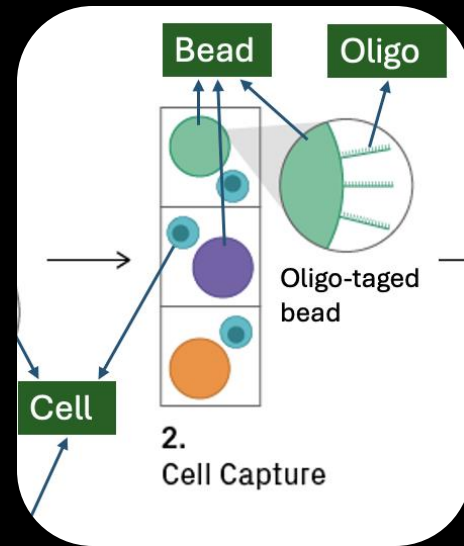


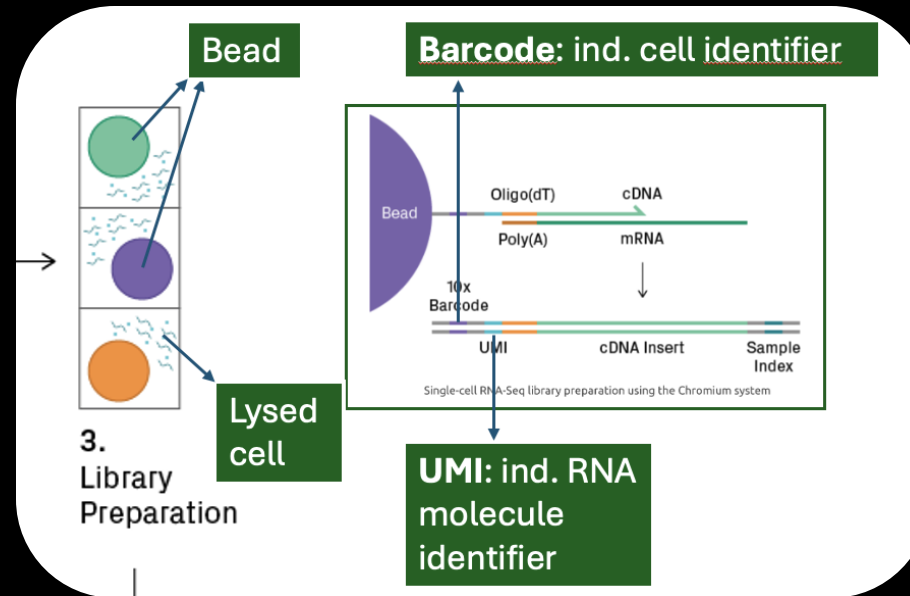


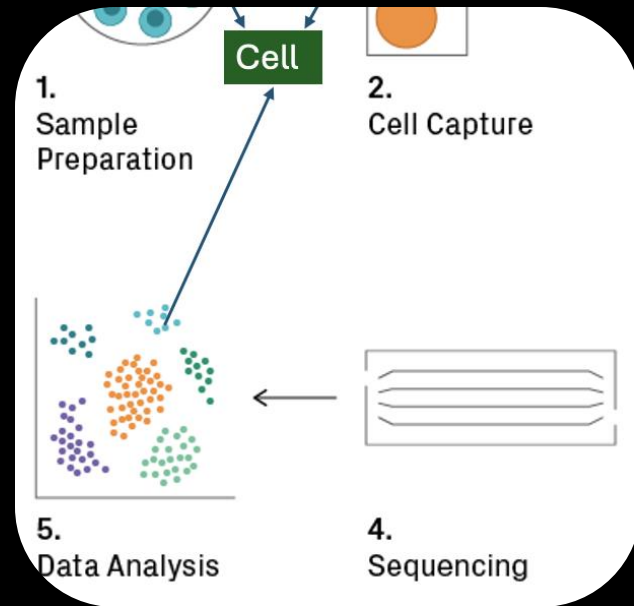


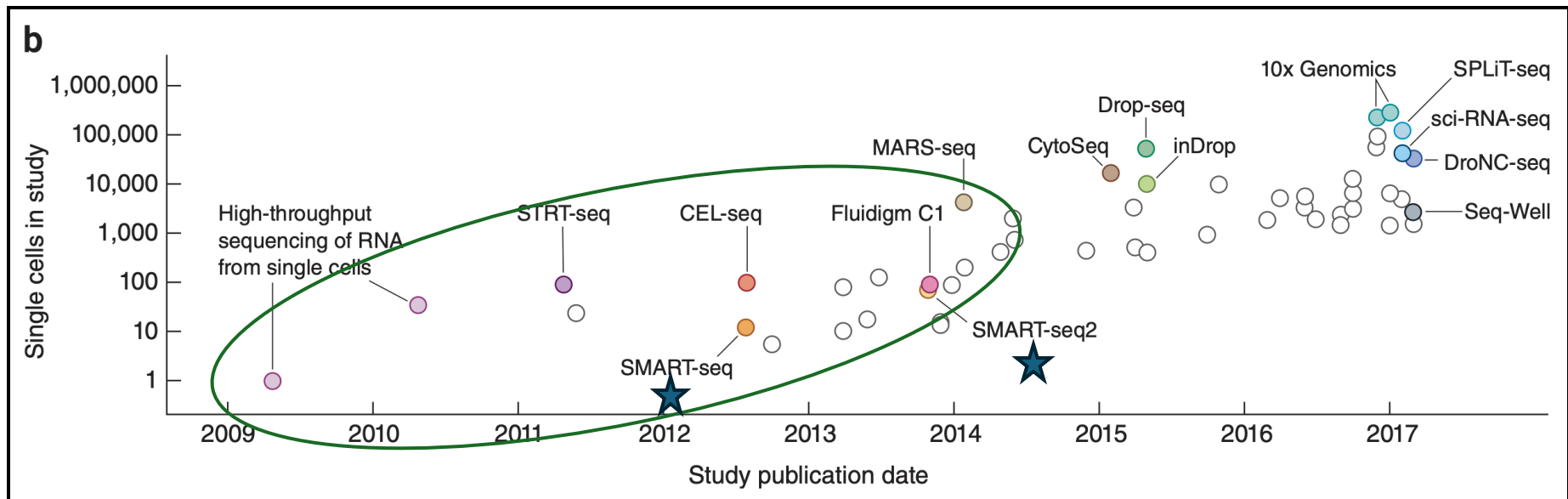




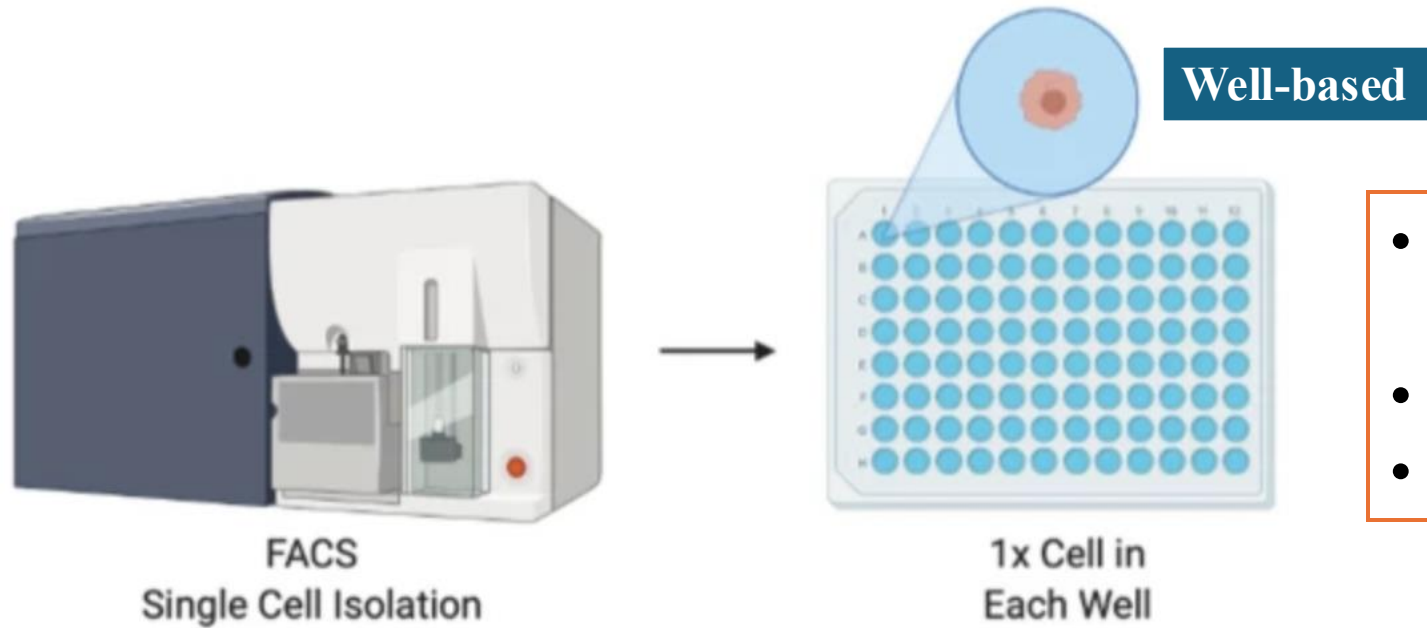






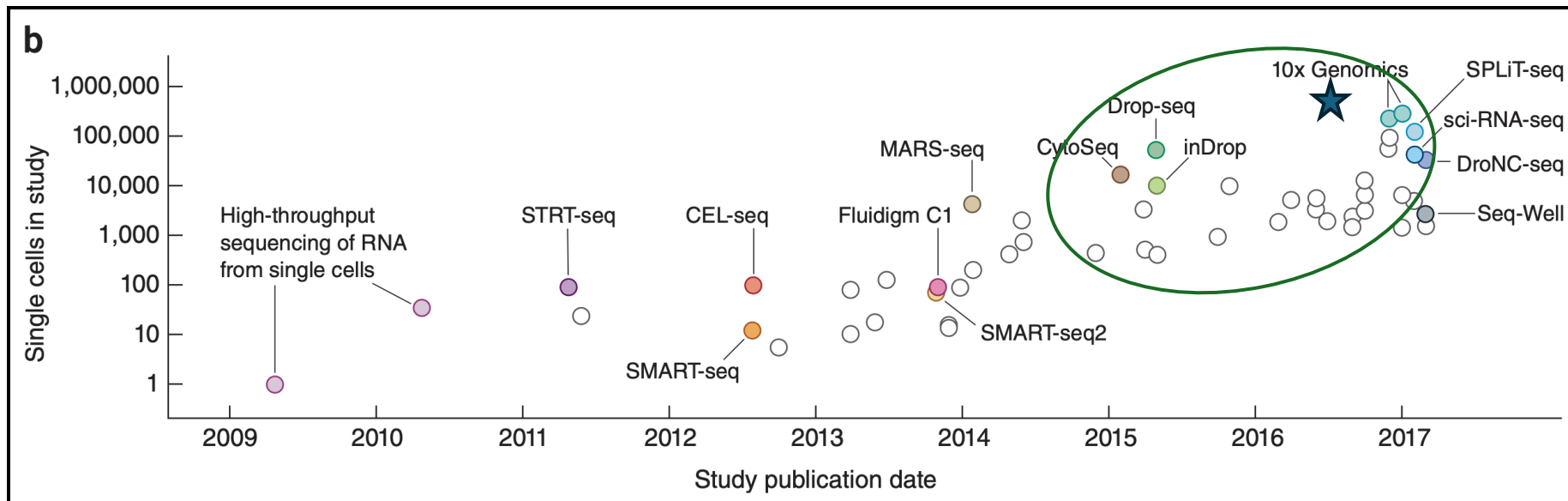


SMART-seq

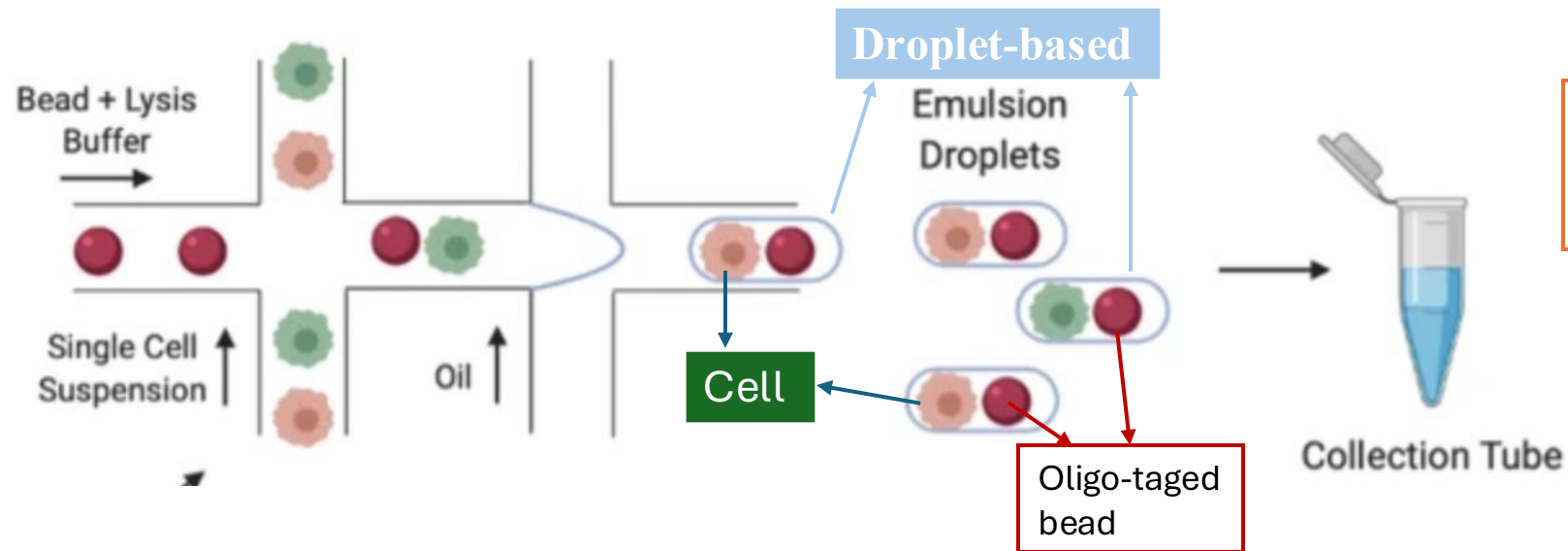


- RNA full length sequencing /seq. deeper (isoform...)
- Less cells
- Lots of manual efforts

Flow cytometry



10X Genomics



- Lots of missing values
- A large number of cells

Microfluidic device

Pair discussion

- Think about a situation where scRNAseq is better than bulk RNAseq

Learning goals

- ❖ What is scRNAseq?
- ❖ How it works?
- ❖ The evolution of scRNAseq?
- ❖ The difference between bulk RNAseq and scRNAseq?

